

Trader Performance vs Market Sentiment Analysis

Objective

Analyze the relationship between Bitcoin market sentiment (Fear/Greed) and trader behavior on Hyperliquid.

Data Preparation

- Trader Dataset: 211,224 rows, 16 columns
- Fear & Greed Dataset: 2,644 rows, 4 columns
- No missing values
- No duplicate records
- Datasets merged on daily date after timestamp conversion

1. Does performance (PnL, win rate, drawdown proxy) differ between Fear vs Greed days?

Evidence

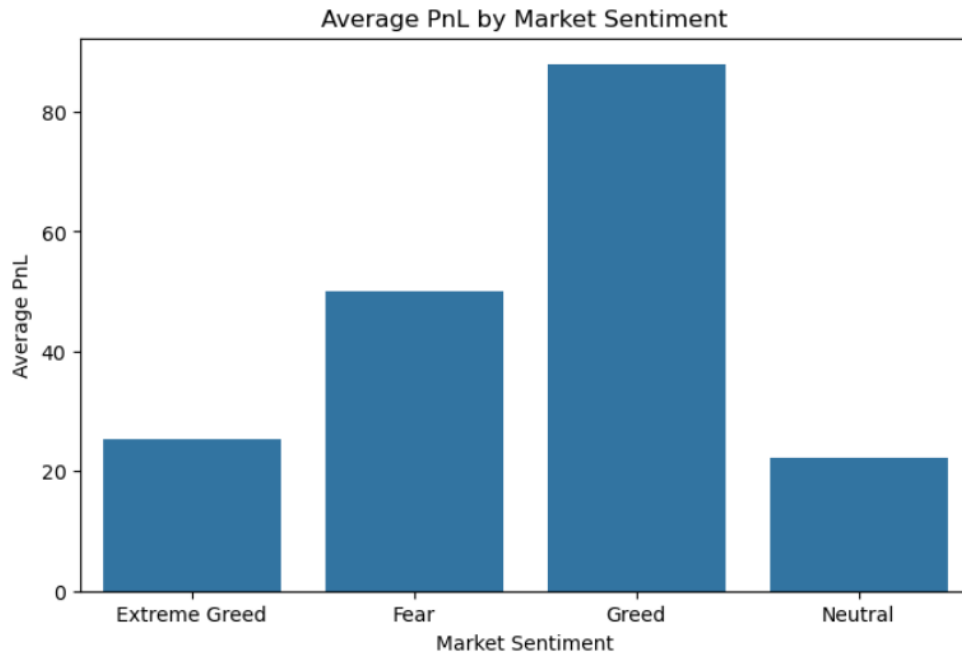
Sentiment	Average PnL	Total PnL
Fear	50.05	6.70M
Greed	87.89	3.19M
Neutral	22.23	0.16M
Extreme Greed	25.42	0.18M

Finding

Greed days generated the highest average profit per trade (87.89), while Fear days generated lower average profitability (50.05).

Interpretation

Traders appear to perform better during optimistic market conditions.



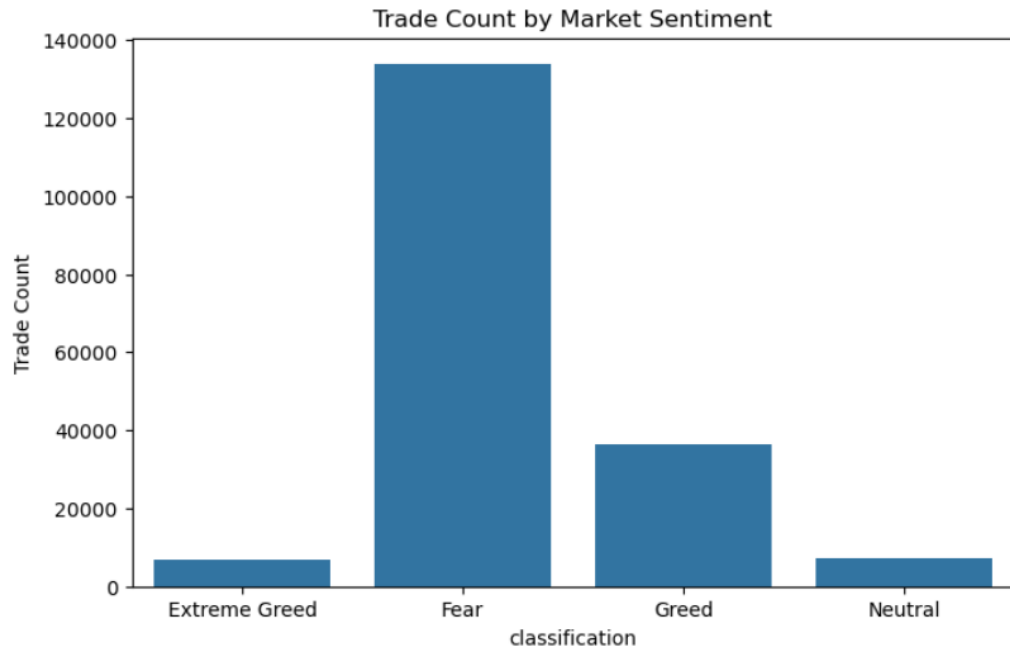
2. Do traders change behavior based on sentiment?

Trade Frequency

Sentiment	Trade Count
Fear	133,871
Greed	36,289
Neutral	7,141
Extreme Greed	6,962

Finding:

Trading activity is significantly higher during Fear periods.



Position Size

Sentiment	Avg Trade Size (USD)
Extreme Greed	5,660
Fear	5,260
Greed	3,183
Neutral	3,059

Finding:

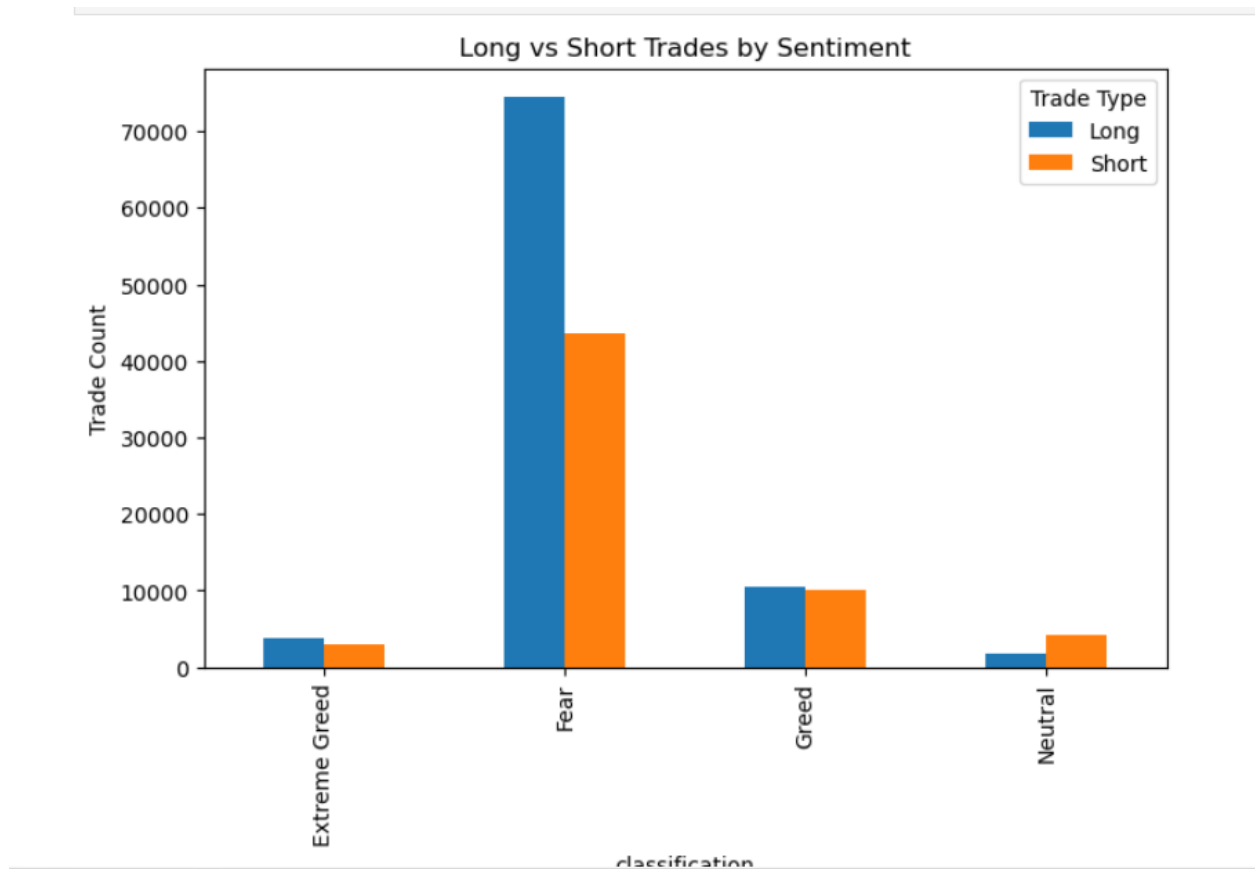
Traders take larger positions during Fear and Extreme Greed periods.

Long/Short Bias

Sentiment	Long	Short
Fear	74,462	43,704
Greed	10,596	10,015
Extreme Greed	3,754	3,045

Finding:

Long positions dominate most market conditions.



3. Trader Segmentation

Segment A: Frequent vs Infrequent Traders

Segment	Avg PnL
Frequent	496,528
Infrequent	147,032

Finding:

Frequent traders generated substantially higher profits.

Segment B: Large vs Small Position Traders

Segment	Avg PnL
Large Position	416,806
Small Position	226,754

Finding:

Larger position traders earned higher average profits.

Segment C: Winners vs Losers

Segment	Avg Trade Size
Winners	6,187
Losers	4,276

Finding:

Winning traders generally used larger average position sizes.

4. Key Insights

1. Greed periods produced the highest average profitability.
2. Fear periods generated the highest trading activity.
3. Larger position traders achieved higher average profits.
4. Frequent traders significantly outperformed infrequent traders.

Strategy Recommendations

1. Increase exposure during Greed periods while maintaining risk controls.
2. Avoid excessive trading during Fear periods.
3. Optimize position sizing rather than simply increasing trade frequency.

Limitations

Leverage information was not available in the dataset; therefore leverage-based analysis could not be performed.

Conclusion

The analysis indicates that market sentiment significantly influences trader behavior and profitability. Greed periods were associated with higher average profits, while Fear periods generated the highest trading activity and larger position sizes. Trader segmentation revealed that frequent traders and large-position traders achieved superior profitability, suggesting that participation level and position sizing play an important role in trading outcomes.